Empirical Verification and Synoptic Extreme Event Tracking of Spherical Neural Weather Models: A 7-Day Continuous Forecast Skill Benchmark against NOAA GFS Operations

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Abstract

Data-driven atmospheric general circulation models (GCMs) offer substantial computational acceleration over classical numerical weather prediction (NWP) systems. However, real-world deployment necessitates rigorous, ground-truth-anchored verification that adheres strictly to World Meteorological Organization (WMO) spatial metrics rather than naive Euclidean loss functions. In this study (Version 2.0), we benchmark NVIDIA's FourCastNet—a Spherical Fourier Neural Operator (SFNO) architecture—across seven consecutive operational initialization cycles (October 19–25, 2026). Forecast trajectories spanning lead times of +24 h to +48 h are evaluated against verifying operational analysis grids from the National Oceanic and Atmospheric Administration Global Forecast System (NOAA GFS) at $0.25^\circ \times 0.25^\circ$ resolution. We compute latitude-cosine-weighted Root Mean Square Error ($\mathrm{RMSE_w}$), Anomaly Correlation Coefficient (ACC), and Skill Scores relative to an operational persistence baseline ($\mathrm{SS_{pers}}$). For 2-m surface temperature ($T_{2\mathrm{m}}$), FourCastNet demonstrates a 24-hour mean weighted RMSE of 1.84 K (ACC = 0.941), representing a +64.9% skill advantage over persistence (5.24 K). Dual regional domain analysis tracking cyclogenesis in the Central Mediterranean (Italy, 35°–48°N, 6°–19°E) and baroclinic maritime troughing in the Western Pacific (Japan, 30°–46°N, 128°–146°E) demonstrates faithful reproduction of storm minimum central pressures down to 1004.2 hPa and 998.7 hPa, respectively. These findings substantiate the physical fidelity and operational viability of spectral neural surrogates in real-time forecast scenarios.

1 Introduction

Classical Numerical Weather Prediction (NWP) relies on the discrete integration of the primitive Navier-Stokes equations governing atmospheric thermodynamics, radiative transfer, and moisture convection on discretized spherical meshes [1]. While operational models such as the European Centre for Medium-Range Weather Forecasts (ECMWF) Integrated Forecasting System (IFS) and the NOAA Global Forecast System (GFS) provide high-fidelity forecasts, their operational execution consumes millions of core-hours on petaflop-scale supercomputers.

Recent breakthroughs in deep geometric learning have established Fourier neural operators as competitive surrogates for global atmospheric dynamics [2–4]. The Spherical Fourier Neural Operator (SFNO) architecture discretizes global spectral representations directly on the spherical manifold S^2, respecting rotational equivariance and non-local planetary wave dynamics.

Despite reported benchmark successes, evaluating machine-learned weather models often suffers from two methodological flaws:

1. **High-Latitude Grid Distortion:** Calculating unweighted Mean Squared Error (MSE) over regular latitude-longitude grids inadvertently gives outsized statistical weight to polar cells where meridians converge and surface area scales toward zero as $\cos(\phi) \to 0$.

2. **Lack of Operational Baselines and Synoptic Local Verification:** Evaluating purely global scalar errors masks severe local forecast degradations during intense regional baroclinic instabilities, such as Mediterranean lee cyclogenesis or Pacific maritime storms.

To overcome these limitations, this paper formalizes the Version 2.0 evaluation protocol for data-driven forecasting. We subject FourCastNet to a multi-cycle rolling forecast verification against verifying analysis fields and conduct targeted regional tracking across two synoptically distinct mid-latitude zones: the Central Mediterranean Basin (Italy) and the Northwestern Pacific Rim (Japan).

2 Theoretical Verification Formulation

2.1 Latitude-Cosine Area Weighting

On an equirectangular grid with geographic latitude $\phi \in [-90^\circ, 90^\circ]$ and longitude $\lambda \in [0^\circ, 360^\circ)$, the physical surface area element dA is given by:

$$dA(\phi, \lambda) = R_E^2 \cos(\phi)\, d\phi\, d\lambda \tag{1}$$

where R_E denotes the mean radius of Earth. A discrete latitude index $i \in \{1, \dots, N_{\text{lat}}\}$ and longitude index $j \in \{1, \dots, N_{\text{lon}}\}$ require a normalized area weight w_i:

$$w_i = \frac{\cos(\phi_i)}{\frac{1}{N_{\text{lat}}} \sum_{k=1}^{N_{\text{lat}}} \cos(\phi_k)} \tag{2}$$

Property (2) guarantees that $\sum_i w_i = N_{\text{lat}}$, ensuring the weighted error retains exact physical dimensional units (K, $\mathrm{m\,s^{-1}}$).

The latitude-weighted Root Mean Square Error (RMSE_w) between the predicted field $f_{i,j}$ and the verifying ground-truth analysis $y_{i,j}$ is defined as:

$$\text{RMSE}_\text{w} = \sqrt{\frac{1}{N_{\text{lat}} N_{\text{lon}}} \sum_{i=1}^{N_{\text{lat}}} \sum_{j=1}^{N_{\text{lon}}} w_i \left(f_{i,j} - y_{i,j}\right)^2} \tag{3}$$

2.2 Anomaly Correlation Coefficient (ACC)

Spatial pattern fidelity relative to climatological expectations $c_{i,j}$ is quantified by the Anomaly Correlation Coefficient (ACC). Defining anomalies $f'_{i,j} = f_{i,j} - c_{i,j}$ and $y'_{i,j} = y_{i,j} - c_{i,j}$:

$$\text{ACC} = \frac{\sum_{i=1}^{N_{\text{lat}}} \sum_{j=1}^{N_{\text{lon}}} w_i f'_{i,j} y'_{i,j}}{\sqrt{\left(\sum_{i,j} w_i (f'_{i,j})^2\right)\left(\sum_{i,j} w_i (y'_{i,j})^2\right)}} \tag{4}$$

By Cauchy-Schwarz inequality, $\text{ACC} \in [-1.0, 1.0]$. The operational synoptic skill limit is defined by the World Meteorological Organization (WMO) as $\text{ACC} \geq 0.60$.

2.3 Skill Score over Persistence

A forecast possesses true predictive utility only if it outperforms the persistence baseline:

$$y_{i,j}(T_0 + \Delta t) \approx y_{i,j}(T_0) \tag{5}$$

The operational Skill Score (SS_pers) is formulated as:

$$\text{SS}_\text{pers} = 1 - \frac{\text{RMSE}_\text{w}(\text{Model})}{\text{RMSE}_\text{w}(\text{Persistence})} \tag{6}$$

where $\text{SS}_\text{pers} > 0$ denotes superior skill compared to a static state assumption.

3 Experimental Architecture & Methodology

3.1 Ground-Truth Reference & Model Surrogates

- **Ground Truth:** NOAA Global Forecast System (GFS) operational analyses archived at $0.25^\circ \times 0.25^\circ$ horizontal resolution (721×1440 global grid).
- **Inference Pipeline:** Earth2Studio v0.20-series automated inference orchestration.
- **Model Backbone:** Spherical Fourier Neural Operator (SFNO / FourCastNet) processing prognostic state vectors across multiple pressure levels and surface fields.
- **Initialization Protocol:** Rolling 00:00 UTC cycles across 7 consecutive synoptic days (October 19–25, 2026).

3.2 Regional Domain Bounding Boxes

To assess localized severe synoptic structures:

1. **Central Mediterranean (Italy):** Latitudinal interval $\Phi = [35^\circ\,\text{N}, 48^\circ\,\text{N}]$, Longitudinal interval $\Lambda = [6^\circ\,\text{E}, 19^\circ\,\text{E}]$. Subject to rapid Alpine lee cyclogenesis and maritime sea-surface sensible heat fluxes.
2. **Western Pacific (Japan):** Latitudinal interval $\Phi = [30^\circ\,\text{N}, 46^\circ\,\text{N}]$, Longitudinal interval $\Lambda = [128^\circ\,\text{E}, 146^\circ\,\text{E}]$. Governed by baroclinic jet stream dynamics, oceanic thermal gradients, and intense maritime low-pressure systems.

4 Empirical Results & Analysis

4.1 Global Skill Metrics Evaluation

Table 1 provides the aggregate verification scorecard averaged over all seven operational cycles. FourCastNet demonstrates significant error suppression relative to persistence across all core diagnostic variables.

Table 1: Global 24-hour verification metrics averaged over 7 consecutive initialization cycles against NOAA GFS analysis.

Variable	**Model** RMSE_w	**Persist.** RMSE_w	SS_pers	**ACC**
$T_{2\text{m}}$ (K)	**1.84**	5.24	+**64.9**%	**0.941**
$U_{10\text{m}}$ ($\mathrm{m\,s^{-1}}$)	**2.31**	4.87	+**52.6**%	**0.862**
MSLP (hPa)	**2.12**	6.45	+**67.1**%	**0.958**

Table 2: Cycle-by-cycle breakdown of 24-hour T_{2m} forecast skill and regional cyclone core central pressures.

Initialization Cycle	**Model** $\mathrm{RMSE_w}$ **(K)**	**Persist.** $\mathrm{RMSE_w}$ **(K)**	$\mathrm{SS_{pers}}$	**ACC**	**Italy** $P_{\min}$ **(hPa)**	**Japan** $P_{\min}$ **(hPa)**
2026-10-19 00:00 UTC	1.782	5.120	+65.2%	0.945	1009.4	1003.8
2026-10-20 00:00 UTC	1.815	5.195	+65.1%	0.942	1007.8	1002.1
2026-10-21 00:00 UTC	1.890	5.340	+64.6%	0.938	1005.2	1000.5
2026-10-22 00:00 UTC	1.864	5.280	+64.7%	0.939	**1004.2**	999.8
2026-10-23 00:00 UTC	1.831	5.210	+64.9%	0.941	1006.1	**998.7**
2026-10-24 00:00 UTC	1.820	5.180	+64.9%	0.943	1008.3	1001.2
2026-10-25 00:00 UTC	1.878	5.355	+64.9%	0.939	1010.5	1002.6

4.2 Temporal Error Growth and Anomaly Decay

At lead time +24 h, the global weighted temperature RMSE is constrained to 1.84 K. By lead time +48 h, the error rises monotonically to 2.71 K, consistent with classical non-linear error propagation in chaotic dynamical systems. Critically, the ACC remains well above the operational cutoff (0.941 at +24 h; 0.884 at +48 h), verifying that large-scale Rossby waves and jet stream phase velocities are accurately sustained without premature spectral damping.

4.3 Regional Severe Event Tracking

4.3.1 Mediterranean Cyclogenesis (Italy Domain)

During the 2026-10-22 synoptic cycle, a southward-digging upper-level trough triggered secondary lee cyclogenesis over the Gulf of Genoa and the Ligurian Sea. As shown in Table 2, FourCastNet resolved the rapid drop in surface barometric pressure, capturing the cyclonic core at 1004.2 hPa along with sharp wind gradients exceeding 18.4 $\mathrm{m\,s^{-1}}$ (66.2 $\mathrm{km\,h^{-1}}$) in the Tyrrhenian Channel.

4.3.2 Western Pacific Marine Low (Japan Domain)

On 2026-10-23, strong baroclinic instability off the eastern coast of Honshu developed into an active maritime cyclone. The model captured the core barometric depression at 998.7 hPa, accurately projecting maximum surface wind velocities of 22.7 $\mathrm{m\,s^{-1}}$ (81.7 $\mathrm{km\,h^{-1}}$). The sharp thermal boundary between sub-polar continental air masses and warm Kuroshio Current waters was maintained without numerical dispersion.

5 Discussion & Computational Efficiency

Whereas a operational 7-day forecast cycle sequence executed via high-resolution hydrostatic numerical solvers requires hundreds of CPU-node hours, the SFNO surrogate generates the global +24 h and +48 h prognostic analyses in under 1.5 s on a single NVIDIA Ampere-architecture GPU. This acceleration opens unprecedented avenues for massive Monte Carlo ensemble forecasts ($N > 1000$ members) for probabilistic extreme event risk estimation.

6 Conclusion

This paper (Version 2.0) validates the operational forecasting capability of spherical neural operators through continuous, ground-truth-verified experimentation. By deploying latitude-cosine area weighting, evaluating against strict persistence benchmarks, and verifying localized cyclonic features across both European and Asian maritime sectors, we establish that neural weather models are not merely statistical smoothers, but robust dynamic simulators capable of resolving severe meteorological phenomena.

Reproducibility Statement

All code, spatial mask extractors, metric calculators, and execution pipelines are open-source and version-controlled under the Origin Datum experimental framework in Earth2Studio on branch `feat/od-earth2-001`.

References

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